

A 12 to 1 V Five Phase Interleaving GaN POL Converter for High Current Low Voltage Applications

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Abstract—As applications like high performance cloud computing grow exponentially, there is an immediate need for high efficiency high current point of load (POL) converters. Interleaving phases is a common technique to reduce conduction loss, lower output ripple, improve high load performance, and reduce stress on power switching devices. However, multiphase POLs are primarily limited by size. The development of Gallium Nitride (GaN) power semiconductors is enabling higher power density in power supply applications. This paper demonstrates this with a 120 W five-phase interleaving GaN synchronous buck converter that achieves a peak efficiency above 96% and maintains over 90% efficiency up to 105 A.

I. INTRODUCTION

Modern microprocessors typically operate between 0.9 - 1.2 V and current demand is on the rise [1], [2]. The Point of Load (POL) is often the lowest efficiency power conversion stage due to high conduction loss and low output voltage. Additionally, conduction loss between the POL converter and the load dissipates additional heat near the processor. The solution lies in the development of efficient converter design and small size that can be physically located near the load. Thus, the application of GaN is pivotal to increase the power density and efficiency of high performance switch-mode power supplies [3], [4]. The effectiveness of GaN is further enhanced in buck converter topology with the application of parallel synchronous rectifiers (SR) [5]. One of the primary drawbacks to GaN technologies is the reluctance of the industry to adopt a new unproven device technology over silicon (Si); however, there are promising results after 125 °C reliability tests and over a year of continuous operation [6], [7].

One approach to increasing efficiency is the expansion to multiple phases. This reduces the overall inductor current ripple which improves the reliability of the devices as well as the load that is being driven by the converter [8]. The multiphase converter also increases the effective output switching frequency which results in smaller size and higher power density [9]. Multiphase converters in previous efforts have demonstrated operating temperatures on the power devices of approximately 55 °C up to 60 °C at 40 A (10 A per-phase) output current with thermal management control techniques and forced air flow [10]. The presented design looks to lessen the need for advanced control techniques and minimize the need for forced air.

II. INTERLEAVING

One problem with operating multiple conversion phases in parallel is that if they are driven by the same driving signal then the magnitude of the output voltage ripple increases. This is fixed by even spacing of the PWM signals, called interleaving. The PWM control was implemented on a TI F28335 control-Card which was chosen for its five easily programmable pulse width modulation (PWM) signals.

These five PWM signals are interleaved and switched at 160 kHz as seen in Fig. 1. The resulting current output waveform is seen in Fig. 2. The interleaving results in a reduced output voltage ripple [11]. This can be seen in the steady state results of a single and five phase board. Typical peak to peak values of a single phase range from approximately 0.95 to 1.03 V when operating at 5 A, as seen in Fig. 3. Five phases range from approximately 0.98 to 1.01 V, as seen in Fig. 4. The output voltage ripple decreases from ± 80 mV to ± 32 mV peak to peak due to the interleaving.

The expansion to multiple phases also introduces a new term, called the effective switching frequency, which is the frequency of the output signal which is seen by the load. With multiple phases interleaved, the load sees a higher frequency than each individual phase. The effective switching frequency (F_{eff}) is defined as the switching frequency (F_s) multiplied by the number of active phases (N) Eq. 1. Increasing the frequency reduces the need for addition passives required by the converter.

$$F_{eff} = F_s \times N \quad (1)$$

III. RESULTS

This multiphase POL converter significantly improves high load performance upon commercial state-of-the-art POLs, minimizes output voltage ripple by interleaving phases, and increases output frequency from 800 kHz up to 2 MHz. Each synchronous buck phase is configured with a single GaN EPC2015 switch and two parallel GaN synchronous rectifiers (SRs) located on the same PCB, as seen in Fig. 5. Due to the design operating as a 12 to 1 V converter, the switch is on for roughly 8% of the time and the SR is on for 92% of the time.

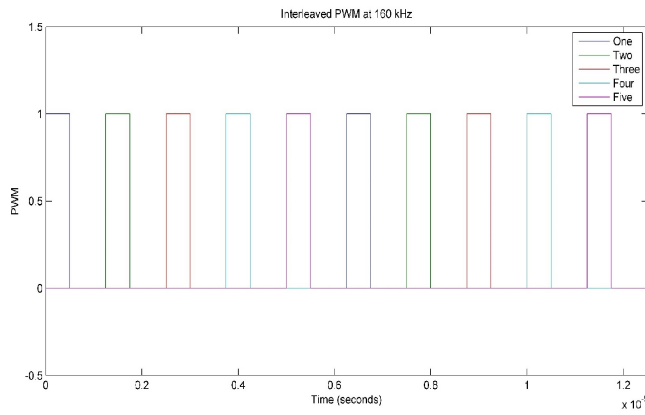


Fig. 1. Five interleaved PWM signals used in the five phase design. $F_s = 160$ kHz $F_{eff} = 800$ kHz

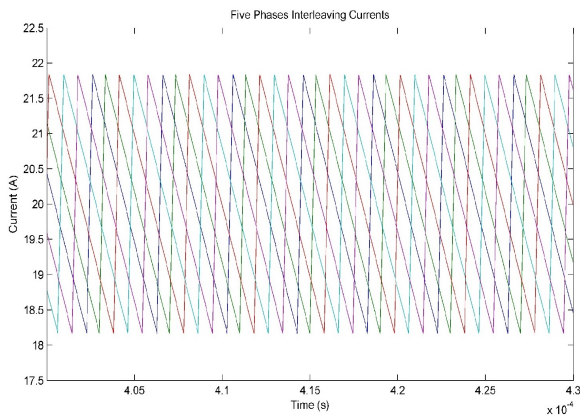


Fig. 2. Five current phases resulting from the interleaving PWM signals.

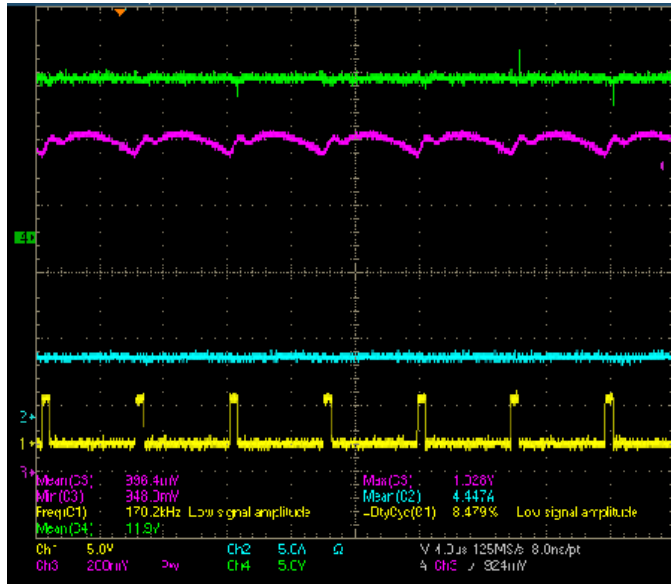


Fig. 3. Steady state operation of single phase at 5 A. V_{OUT} in magenta

To increase the efficiency of the system, two SRs are used in parallel since this reduces the on-resistance of the combined SR which conduct a majority of the time.

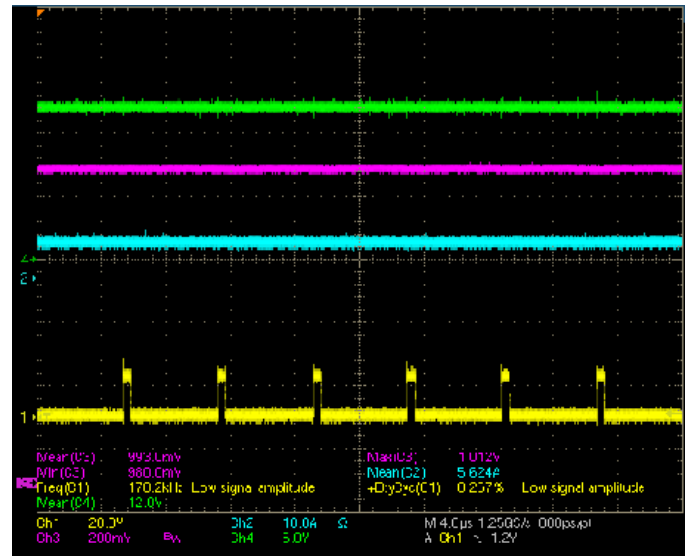


Fig. 4. Steady state operation of five phases at 5 A. V_{OUT} in magenta

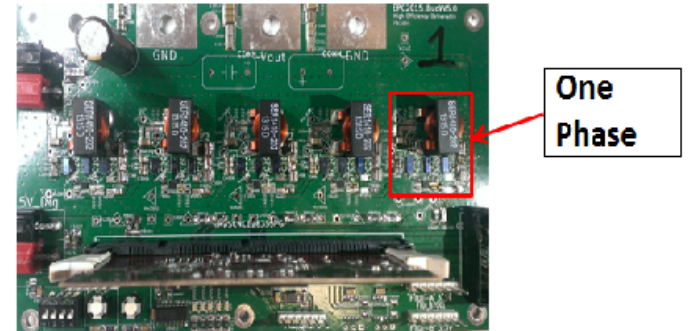


Fig. 5. The five phase synchronous buck converter using GaN HEMTs with peak efficiency over 96%.

The POL is designed with a wide input and output operating range: $V_{IN} = 2 - 24$ V and $V_{OUT} = 0.5 - 5$ V. With removal of developmental test points, the converter has a power density of approximately 270 W/in^2 , a feat enabled by GaN power semiconductors.

With five phases sharing the load current, the heat produced next to the load can be decreased, which decreases the reliance on external cooling. Thermal images of a single and multiphase converter are taken with the same output current of 20 A. The current sharing between the five phase converter means that each individual phase is effectively operating at 4 A each. Both of these designs have the same individual buck layout with only the phase count increasing and no forced air cooling. The single phase results can be seen in Fig. 6 and the five phase results in Fig. 7. Here a 18.4°C temperature difference can be seen where the single phase peaks at 46.6°C and the five phase at 28.2°C on the GaN power switching device.

The five phase design also showed superior thermal performance when each phase carried the same amount of current. The single phase board operating at 10 A output current reached a peak temperature of 37.6°C Fig. 8. The five phase board operating with 10 A per-phase (50 A total) peaked at only 31.9°C Fig. 9. This shows a 5.7°C improvement

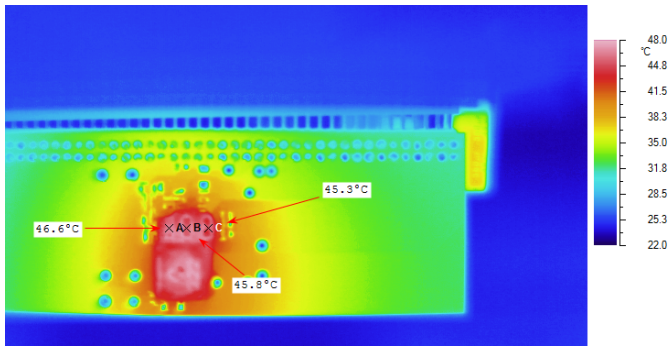


Fig. 6. Thermal image of a single phase converter operating at 20 A output current showing high heat dissipation near the load

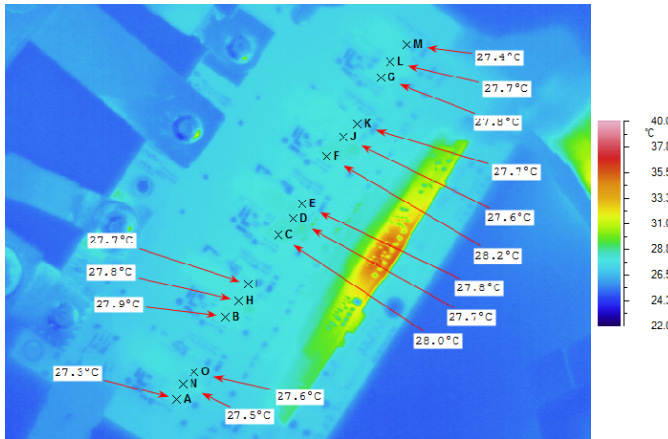


Fig. 7. Thermal image of a five phase converter operating at 20 A output current showing the reduced heat dissipation near the load

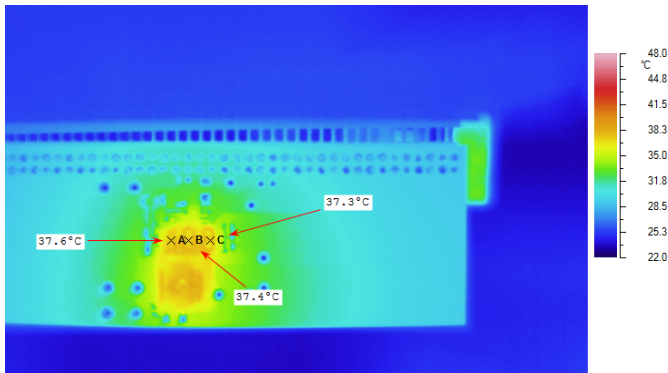


Fig. 8. Thermal image of the single phase converter operating at 10 A load and no forced air cooling and a peak temperature of 37.6°C

in the five phase design over the single phase design for comparable operating currents per-phase. Although this design lacks complex thermal management controller techniques, the difference in temperatures between the center and outside GaN switches is only 2°C at 50 A due in part to an optimized layout.

Various switching frequencies were tested where 160 kHz was determined to be most efficient for the five phase converter. The POL achieves a peak efficiency over 96% at 25 A output current. While operating at 25 A, the converter never gets

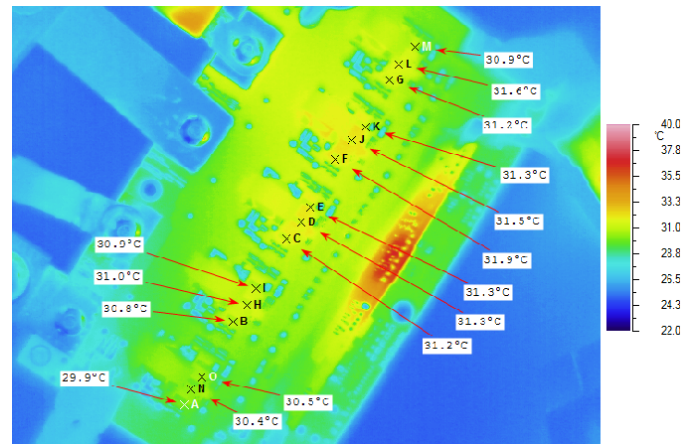


Fig. 9. Thermal image of the five phase converter operating at 50 A load and no forced air cooling and a peak temperature of 31.9°C

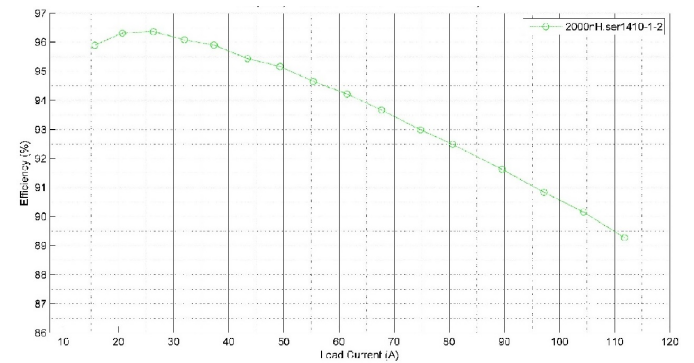


Fig. 10. Load vs efficiency for the five phase converter which maintains over 90% efficiency up to 105 A. $F_s = 160$ kHz $F_{eff} = 800$ kHz

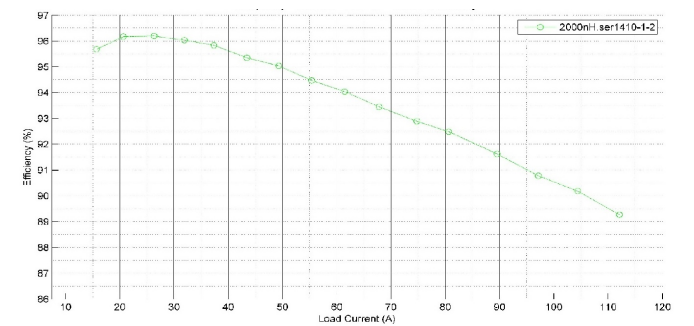


Fig. 11. Load vs efficiency for the five phase converter which maintains over 90% efficiency up to 100 A. $F_s = 200$ kHz $F_{eff} = 1$ MHz

over 28.4°C, minimizing the need for external cooling. The converter maintains this high performance with over 90% efficiency up to 105 A, as seen in Fig. 10. Although the system peaks at an effective switching frequency of 800 kHz, the system still performs well at an effective output switching frequency of 1 MHz maintaining over 96% efficiency over 30 A and maintains over 90% efficiency over 100 A Fig. 11. Increasing the effective switching frequency to 2 MHz, the system peaks efficiency peaks over 94% and maintains over 90% efficiency up to 95 A, as seen in Fig. 12.

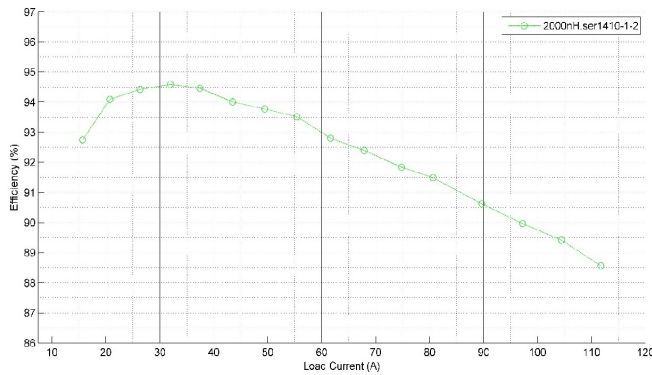


Fig. 12. Load vs efficiency for the five phase converter which maintains over 90% efficiency up to 95 A. $F_s = 400$ kHz $F_{eff} = 2$ MHz

IV. CONCLUSION

A five phase synchronous buck converter has been developed to maximize efficiency for modern microprocessors that demand low voltage and high current operation. A new term for multiphase converters, called the effective switching frequency was introduced describing the frequency seen by the load. This results in a significantly reduced output voltage ripple as well as optimizing efficiency at higher loads.

The presented POL reaches a peak efficiency of 96% at 25 A and maintains over 90% efficiency up to 105 A output current with an effective switching frequency of 800 kHz. This design also performs well at an effective switching frequency, of 1 MHz and 2 MHz with over 90% efficiency maintained over 100 A and 95 A, respectively. The 120 W design features two parallel SR GaN power devices for each individual phase to minimize the on-resistance.

A detailed thermal analysis was performed on both a single and five phase design. The converter showed superior thermal performance over the single phase design with a 5.7 °C lower operating temperature at a comparable 10 A per-phase output current. The five phase converter design minimizes the heat dissipation near the load. All of the thermal results presented are without any external forced air cooling.

Overall, the 12-1 V five phase converter has demonstrated the effectiveness of GaN used in switched mode power converters. This design boasts over 96% efficiency, and an effective switching frequency up to 2 MHz which results in minimal heat dissipation. With GaN still considered a young technology, this leaves room for improvements of future designs.

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